

Akshay Hinduja
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Education: Ph.D. Robotics, Dept. of Mechanical Engineering, Carnegie Mellon University

Advisor : Dr. Michael Kaess

July 2024 | Pittsburgh, PA

- M.S. Robotics, Dept. of Mechanical Engineering, Carnegie Mellon University

Advisor: Dr. Kenji Shimada

May 2017 | Pittsburgh, PA

- B.Tech Production Engineering, VJTI

May 2014 | Mumbai, India

Expertise:

- Computer vision, and vision based deep learning algorithms
- Simultaneous Localization and Mapping (SLAM) algorithms
- Underwater robotics, acoustic localization and sonar perception
- Sensor fusion
- Driver development

Professional Experience:

- **MBARI: Software Engineer, Embedded Data and Algorithms**

July 2024 – Current | Moss Landing, CA

- Driver development for multicamera systems on edge computers
 - Wrote custom drivers developed for multicamera systems utilizing multithreading techniques for 2k@60fps synchronous or asynchronous data streaming and recording.
- Embedded AI and Vision systems engineering
 - Wrote python code to interface with optical sensors and use data streams for computer vision tasks such as tracking of marine animals.
 - Develop and refine machine learning models for use on embedded systems for on-device processing
 - Interface backseat computing with frontseat commands to enable reactive vehicle behaviors.
- Driver development for acoustic sensors
 - Wrote python and c++ drivers to interface with wide band echo sounders used on MBARI's remotely operated vehicles.
 - Wrote visual and CLI tools to interact with drivers to change parameters and visualize returns over low-bandwidth and sparse wireless communications.
- Localization in GPS denied environments using bathymetric submaps and prior surveys
 - Utilized Stochastic Variational Gaussian Processes to help localize AUVs in low altitude surveys when USBL lock is unavailable.
 - Develop a fusion system tying Particle Filter updates to a pose graph to incorporate submap based loop closures to generate accurate and geo-localized 3D maps and trajectory estimates.
- 3D log playback
 - Wrote ROS2 and RViz docker images to ingest vehicle logs and playback 3D representations of navigation during dives, including readouts of vehicle state data and science readings.
- Terrain Avoidance and Following:

- Wrote algorithms to use profiling imaging sonars and DVL input to maintain altitude and correct for impending obstacles using forward looking horizon modeling.
- **Robot Perception Lab: Robotics Systems Engineer/Extern**
June 2017 – June 2018 | Carnegie Mellon University, Pittsburgh
 - Setup and maintenance of software stack for multi-sensor pack used for recording data
 - Stereo vision-based dense reconstruction and damage detection on rooftops. Using a combination of statistical feature descriptors and ML, a 3D classifier was developed to detect minor geometric anomalies.
 - Developed a feature-based SLAM technique using imaging sonars. Used A-KAZE features to perform real-time SLAM on simulated and real-world datasets.
- **Tsuneishi Shipbuilding Co Ltd.: Robotics Intern**
May 2016 – June 2016 | Fukuyama, Japan
 - Robot localization using Velodyne VLP 16 Lidar. Implemented ICP and NDT-based localization using pose updates for 3D pointcloud data.

Projects:

- **Pose-supervised Learning for Imaging Sonars**
Ph.D. thesis research | Robot Perception Lab | 2023, Pittsburgh, PA
 - Developed a pose supervised network to learn viewpoint invariant feature correspondence between imaging sonar images of the same scene.
- **Using cGANs To Filter Imaging Sonar Data for Occupancy Mapping**
Ph.D. research | Robot Perception Lab | 2022, Pittsburgh, PA
 - Use cGANs to train a network to recognize speckle noise in sonar images and to filter the image to give clean and robust information for generating accurate occupancy maps.
- **Acoustic Localization and Communication Techniques for Multiple, Low Power, Low-Cost Soft Robots**
Ph.D. thesis research | Robot Perception Lab | 2021, Pittsburgh, PA
 - Develop acoustic techniques to localize and direct several soft robotic fish using a known network of speakers on the transmission side and a single MEMS Microphone on each robot agent.
- **Soft-Sensor Integration for Information Rich Mapping**
Ph.D. thesis research | Robot Perception Lab | 2020, Pittsburgh, PA
 - Integrating custom “soft sensor” stickers onto existing underwater vehicles to obtain data pertaining to chemical properties and other modalities to integrate into the existing pipeline of 3D Map generation.
- **Degeneracy Aware Mapping for Underwater Robots**
Ph.D. thesis research | Robot Perception Lab | 2019, Pittsburgh, PA
 - Developing SLAM optimization improvements for degenerate environments. Focus on methods to aid robots with low-quality sensors.
- **Development Of A Perception System For A Voice Controlled Husky**
Robot Autonomy | Field Robotics Center | Spring 2017, Pittsburgh, PA
 - Development of a human detection and tracking package in ROS using an Adonis camera, Velodyne Lidar, and Hokoyu line scanner. Developed for the ClearPath Husky mobile robot platform.
- **3D Point Cloud NDT Odometry**
Computer Vision | Fall 2016, Pittsburgh, PA
 - Developed a visual odometry package in ROS using a Velodyne Lidar VLP16. The algorithm was based on the use of the Normal Distribution Transform for scan matching.

- **Computational Engineering and Research Lab: Graduate Research Student**

Master's Thesis | August 2015 – May 2017 | Pittsburgh, PA

- Designed and built a prototype for an autonomous robot to weld ship hulls for commercial merchant vessels and tested the suitability of the design for its movements across the environment.
- Developed a frame-to-frame, real-time odometry estimator using the Normal Distribution Transform and a Velodyne VLP-16

Publications:

- "Cross-SONIC: Cross-Sensor and Multi-Modal Sonar Image Correspondence" - **A Hinduja**: 2024 PhD Thesis: Underwater Localization and Mapping for Cost-Effective Robots, Carnegie Mellon University.
- "SONIC: Sonar Image Correspondence using Pose Supervised Learning for Imaging Sonars" - S Gode*, **A Hinduja***, M Kaess: 2024 IEEE International Conference on Robotics and Automation (ICRA)
- "Multi-Radar Inertial Odometry for 3D State Estimation using mmWave Imaging Radar" - JT Huang, R Xu, **A Hinduja**, M Kaess: 2024 IEEE International Conference on Robotics and Automation (ICRA)
- "Conditional GANs for Sonar Image Filtering with Applications to Underwater Occupancy Mapping" - T Lin, **A Hinduja**, M Qadri, M Kaess: 2023 IEEE International Conference on Robotics and Automation (ICRA)
- "Acoustic Localization and Communication Using a MEMS Microphone for Low-cost and Low-power Bio-inspired Underwater Robots" - **A Hinduja**, Y Ohm, J Liao, C Majidi, M Kaess: 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- "Degeneracy-Aware Factors with Applications to Underwater SLAM" - **A Hinduja**, BJ Ho, M Kaess: 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- "Designing a Mobile Robot For Double Hull Welding" - CF Goh, **A Hinduja**, L Zhang, D Ajmani, R Song, K Shimada: 2019 Journal of Mechanisms and Robotics 11 (4).
- "Feature-based SLAM for imaging Sonars" - E Westman, **A Hinduja**, M Kaess: 2018 IEEE International Conference on Robotics and Automation (ICRA)

Skills:

- Programming: Python, C++, MATLAB, ROS, OpenCV, Point Cloud Library (PCL), ISAM/GTSAM, PyTorch, Unreal Engine, TensorFlow, Docker
- Good project management skills, capable of coordinating multiple projects and collaborating effectively with teams.